

## 1 Lieber Frank,

hier habe ich meinen aktuellen Stand zusammengefasst. Am Anfang nenne ich nochmal das Setting und das Lemma 2.0.1, das es zu beweisen gilt. Es folgt eine kurze Anmerkung dazu, warum ich das gerne beweisen möchte und wo die Behauptung zu finden ist.

**Am spannendsten für dich** ist, denke ich, Abschnitt 2: "Where I Stand at the Moment". Dort fasse ich meine bisherigen Ergebnisse zusammen und erkläre meine Probleme. Dabei sind auch querverweise auf den nächsten Abschnitt:

Danach folgt einiges an technischem Material, das die bisherigen Beweise enthält. Das ist vor allem der Vollständigkeit halber im Dokument enthalten.

## 2 The Setting

We work in a compact, riemanian, orientable, smooth Manifold  $(M, g)$  of dimension  $m$  together with a Morse-Smale funktion

$$f : M \rightarrow \mathbb{R} \quad (1)$$

We assume that  $q$  is a critical point of index  $k+1$  and  $p$  is a critical point of index  $k$ . we define  $f(q) = b$  and  $f(p) = a$  and assume that there is no kritikal point in  $f^{-1}(a, b) \subseteq M$ . We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}, \quad (2)$$

and the constants

$$c \in (a, b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big} . \quad (3)$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}, \quad (4)$$

$$L_q := \{x \in N_q | f(x) = c\}, \quad (5)$$

$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}, \quad (6)$$

$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}, \quad (7)$$

and finally:

$$C := N_p \cup N_q, \quad (8)$$

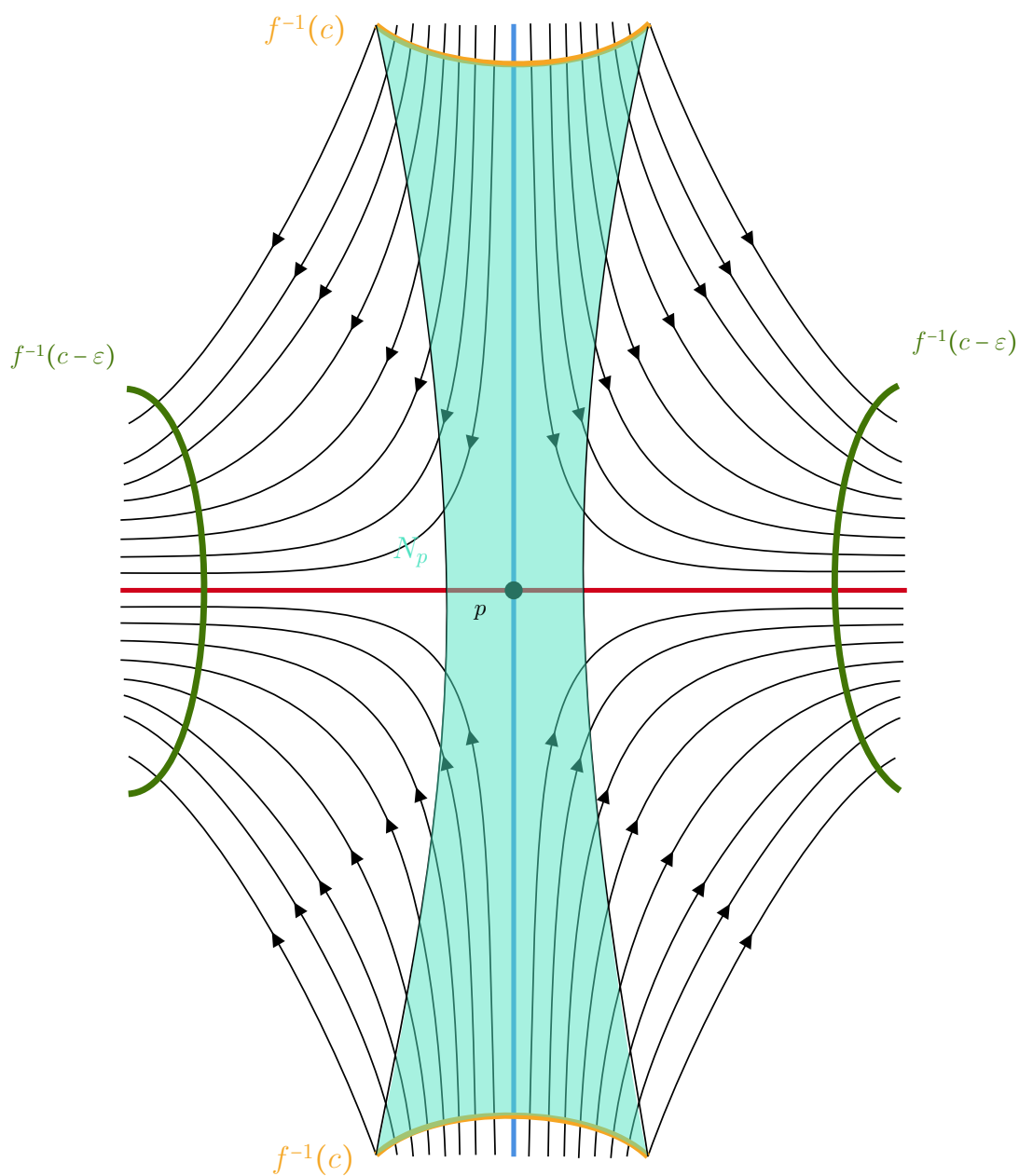
$$B := N_p \cup L_q, \quad (9)$$

$$A := L_p \cup (L_q \overset{\circ}{-} N_p). \quad (10)$$

**Lemma 2.0.1.** *We claim that*

1.  $(N_q, L_q)$  is a regular index pair for  $q$  .
2.  $(C, B)$  is an index pair for  $q$  .
3.  $(N_p, L_p)$  is a regular index pair for  $p$  .
4.  $(B, A)$  is an index pair for  $p$  .
5.  $N_p$  is a tubular neighbourhood of  $W(\rightarrow p) \cap M^c$  . .

**Remark 2.0.2.** For our purpose we don't really need  $N_p$  to be a tubular neighbourhood of  $W(\rightarrow p) \cap M^c$ . What we need is a canonical way to identify the connected components  $V_i$  of the intersection  $N_q \cap N_p = \sqcup_i V_i$  with  $W(p \rightarrow)$ . The tubular neighbourhood tries to argue, that both are fibers in the normal bundle of  $W(\rightarrow p) \cap M^c$  and since the latter is contractible this can be trivialized. In [MorseTheorySalmbon] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [banyaga2004lectures] Banyaga claims this (also without any argument). A Figure below depicts the situation:



### 3 Where I Stand at the Moment

The statements 1-4 are easy to prove by the definitions. The regularity can be derived from a general fact for pairs  $(X, Y)$  in metric spaces: If  $Y$  is closed in  $X$  and there is a neighbourhood of  $Y$  that is open in  $X$  such that  $Y$  is a strong deformation retract of  $U$ . So the only thing left to show is, that  $N_p$  is a tubular neighbourhood. If we have a Morse chart around  $p$ , where the gradient is euclidean, the statement is at least around

$p$  easily done. Compare the proof below (4). (By moving  $c$  close to  $f(p)$  this should then also show the global statement). But there is now way to assume that the gradient is euclidean and the Morse lemma doesn't care for the gradient (it is a statement for smooth and not Riemannian manifolds) Now in general I was able to manipulate the Morse-Chart, such that the gradient at  $p$  is of diagonal form with positiv entries(compare theorem 4.0.7). This gave hope for a Taylor expansion of the flow, since the Jacobean of the flow can be expressed via the gradien. With heavy notation this reads:

$$q_\psi \frac{\partial}{\partial x} \Big|_0 \varphi_t(\psi^{-1}(x)) = \exp \left( \sum_k \left( g^{ki}(0) \frac{\partial^2 (f \circ \psi^{-1})}{\partial x^j \partial x^k} \right)_{ij} \right).$$

Here the left side is the Jacobian of the flow in a chart in the origin. (The notation is unpacked below) Now there are several **problems**:

- Even if  $g^{ij}$  is diagonal and positive definite, it is hard to expand the above expression. whilst maintaining the splitting. compare the current status of that in remark 4.0.8.
- The proof for the "trivial" case where the metric is euclidean relies on a argument that uses local monotomie after a splitting(compare equation 12). This however cannot be derived from a Approximation of the flow.
- The statement of  $N_p$  being a tubular neighbourhood is not enough. Implicitly Salaomon also uses, that the  $V_i$  which can be denoted as the connected components  $f^{-1}(c) \cap N_p$  are fibres in the neighbourhood, or canonically homeomorphic (up to homotopy) to fibres.

A little bit more on the second problem. Assume that we have a two dimensional manifold and in our Morse chart the gradient  $g^{ij}$  for  $i, j \in \{1, 2\}$  is given by the matrix

$$g^{12} = g^{21} = 0, g^{22} = 1 \quad \text{and} \\ g^{11}(x, y) = 1 + e^{\frac{-1}{x^2}} \sin\left(\frac{1}{x^2}\right)$$

This comes from a well defined smooth metric and furthermore in the origin, this is the euclidean. However, the monotony is not controled by the linear approximation. (in fact, this pathological example shows that even the Taylor series wouldn't help). So here I am stuck. This looks like the gradient of the function. I assume that the Morse-Smale condition makes sure that such examples are not possible, but this is something to work on...

## 4 The Technical Parts

*Proof of the Tubular neighbourhood: When the Metric is Euclidean.* For now assume that  $g_{ik} = \text{diag}(1, \dots, 1)$ , i.e. that we work with the euklidean metric. Then the gradient is:

$$\sum_{i,k} \frac{\partial f \circ \psi^{-1}}{\partial x^k} \frac{\partial}{\partial x^i} = 2 \frac{\partial}{x^s} - 2 \frac{\partial}{x^u}.$$

Hence, the flow corresponding to  $\psi_*(-\text{grad}(f))$  is of the form

$$t \mapsto \varphi_t(x) = (e^{-2t}x^s, e^{2t}x^u).$$

Assume that  $\varepsilon$  is small enough, such that all  $x \in \psi(N_p \cap V) \setminus W(\rightarrow p)$  flow through  $f^{-1}(a - \varepsilon)$  inside of  $U$ , i.e. we can formulate the property:

$$f \circ \varphi_T(\psi^{-1}(x^s, x^u)) \geq a - \varepsilon \quad \Leftrightarrow \quad f\left(\psi^{-1}(e^{-2T}x^s, e^{2T}x^u)\right) = a + (e^{-2T}x^s)^2 - (e^{2T}x^u)^2 \geq a - \varepsilon \quad (11)$$

This however reads:

$$a + (e^{-2T}x^s)^2 - (e^{2T}x^u)^2 \geq a - \varepsilon \quad \Leftrightarrow \quad (e^{2T}x^u)^2 \leq \varepsilon + (e^{-2T}x^s)^2 \quad (12)$$

And hence we have that:

$$\psi(N_p \cap V) = \{(x^s, x^u) \in U \mid \|x^u\| \leq R(x^s)\}$$

where  $R(x^s)$  is a smooth function. □

**Lemma 4.0.1.** *Let  $\varphi : U \rightarrow V$  be a chart. Then the gradient has the local form:*

$$\text{grad}(f) = \sum_{i,j} g^{ij} \left( \frac{\partial}{\partial x^i}(f) \right) \frac{\partial}{\partial x^j} =: \sum_{i,j} g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}.$$

Here,  $g^{ij}$  denotes the smooth functions given by inverting the coordinates of the function  $x \mapsto (g_{ij}(x))$  that defines the coordinate representation of the gradient in  $x$ .

*Proof.* Assume that  $v, w \in \Gamma(TM)$  such that  $v = \sum_i v^i \frac{\partial}{\partial x^i}$  and  $w = \sum_i w^i \frac{\partial}{\partial x^i}$ . Then  $g(v, w) = \sum_{i,j} g_{ij} v^i w^j$ . Suppose that  $\text{grad}(f) = \sum_j G^j \frac{\partial}{\partial x^j}$  and  $q$  is the coordinate map  $T\mathbb{R} \rightarrow \mathbb{R}$  in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x^j}(f) = q \circ df\left(\frac{\partial}{\partial x^j}\right) = q \circ \frac{\partial f}{\partial x^j} = g(\text{grad}(f), \frac{\partial}{\partial x^j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned} (G^1, \dots, G^m)(g_{ij}) &= \left( \frac{\partial}{\partial x^1}(f), \dots, \frac{\partial}{\partial x^m}(f) \right) \\ &\Leftrightarrow (G^1, \dots, G^m) = \left( \frac{\partial}{\partial x^1}(f), \dots, \frac{\partial}{\partial x^m}(f) \right) (g^{ij}). \end{aligned}$$

But then we can conclude that  $G^j = \sum_i g^{ij} \frac{\partial}{\partial x^i}(f)$ . □

**Definition 4.0.2** (Jacobian of the Gradient). The gradient is a section into the tangent bundle,  $\text{grad}(f) : M \rightarrow TM$ . Let  $\psi : M \supseteq U \rightarrow V \subseteq \mathbb{R}^m$  be a chart. The coordinate map  $q_\psi$  on  $TU$  assigns to a vector  $v \in T_p M$  (where  $p \in U$  with  $\psi(p) = x$ ) its components with respect to the basis  $\left(\frac{\partial}{\partial x^i}\Big|_p\right)$ , i.e.,  $q_\psi(v) = (v^1, \dots, v^m)$  if  $v = \sum_{i=1}^m v^i \frac{\partial}{\partial x^i}\Big|_p$ . We define the Jacobian of the gradient with respect to the chart  $\psi$  as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\text{grad}(f))_\psi(x) := J\left(q_\psi \circ \text{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial(q_\psi \circ \text{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

**Lemma 4.0.3.** Let  $t \in \mathbb{R}$  be small enough and  $\varphi_t : M \rightarrow M$  be the flow map corresponding to the negative gradient. Assume that  $U$  is the domain of a chart  $\psi$  and  $p \in U$  such that  $\varphi_t(p) \in U$ . Then the linear map  $d\varphi_t|_p : T_p M \rightarrow T_{\varphi_t(p)} M$  has a local representation in the coordinates induced by  $\psi$  given by:

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-q_\psi \circ \text{grad}(f) \psi^{-1} \cdot t\right).$$

*Proof.* For this we consider the function  $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \rightarrow U$ . For fixed  $x$  we can calculate

$$q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) = -q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x} \frac{\partial}{\partial t} \psi \circ \varphi_t(\psi^{-1}(x)) \\ &= \frac{\partial}{\partial x} q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x))) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f) \left( \psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x)) \right) \\ &= -\frac{\partial}{\partial x} \left( q_\psi \circ \text{grad}(f) \circ \psi^{-1} \right) \frac{\partial}{\partial x} \left( \psi \circ \varphi_t(\psi^{-1}(x)) \right) \end{aligned}$$

Hence by the theory of linear differential equation with  $\frac{\partial}{\partial x} \psi \circ \varphi_0(\psi^{-1}(x)) = 1_m$  we have:

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-q_\psi \circ \text{grad}(f) \circ \psi^{-1} \cdot t\right).$$

□

**Lemma 4.0.4.** *Let  $\psi$  be a coordinate system around a critical point  $p$ . I.e.  $\psi : p \in U \rightarrow V$  with  $\psi(p) = 0$  and let  $\left(\frac{\partial}{\partial x_1}|_x, \dots, \frac{\partial}{\partial x_m}|_x\right)$  be the induced basis of  $T_x M$ . Then the jacobian of the gradient reads*

$$J(\text{grad}(f))_\psi(0) = \sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}. \quad (13)$$

*Proof.* Let  $q_\psi$  denote the koordinate funktion  $TU \rightarrow R^m$  induced from the basis  $\left(\frac{\partial}{\partial x_1}|_x, \dots, \frac{\partial}{\partial x_m}|_x\right)$ . then the gradient in  $\psi$  reads:

$$q_\psi \circ \text{grad}(f) = \left( \sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k} \right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} Q \circ \text{grad}(f) \circ \psi^{-1} &= \left( \frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k} \right)_{ij} \\ &= \left( \sum_k \left[ \frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}. \end{aligned}$$

and now in  $p = \psi^{-1}(0)$  we have

$$\frac{\partial}{\partial x} Q \circ \text{grad}(f) \circ \psi^{-1} \Big|_0 = \left( \sum_k \left[ g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}.$$

□

**Theorem 4.0.5** (Sylvesters Law of Inertia). *Let  $A \in \text{Mat}(n, \mathbb{R})$  be symmetric and let  $T, T' \in \text{GL}(n, \mathbb{R})$  and  $k, k', l, l' \in \mathbb{N}$  such that*

$$T^t \circ A \circ T = \begin{pmatrix} 1_k & 0 & 0 \\ 0 & -1_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} 1_{k'} & 0 & 0 \\ 0 & -1_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

*then  $k = k'$ ,  $l = l'$  and  $\text{rank}(A) = k + l$ .*

*Proof.* Since  $T, T'$  are invertible we have that

$$k + l = \text{rank}(T^t \circ A \circ T) = \text{rank}(A) = \text{rank}(T'^t \circ A \circ T') = k' + l'. \quad (14)$$

Hence it suffices to show that  $k = k'$ . Which we will do by proofing the claim:

$$k = \max \left\{ \dim(U) \mid U \subseteq \mathbb{R}^n \text{ subspace such that } x^T A x > 0 \ \forall x \in U \setminus \{0\} \right\}$$

So we start by showing „ $\leq$ “: Denote the first  $k$  colomus of  $T$  with  $x_1, \dots, x_k$ . They form a basis of  $\mathbb{R}^n$  and with  $0 \neq x = \sum_{i=1}^k \lambda_i x_i$  we have that by bilinearity

$$x^T A x = \sum_{i=1}^k \lambda_i x_i^T A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^T A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0. \quad (15)$$

This concludes the first inequality.

Now let  $U$  be any  $k$ -dimensional subspace such that for all non zero  $x \in U$   $x^T A x > 0$ . By a calculation analog to the one above we have that for any  $x \in W := \text{span}(x_{k+1}, \dots, x_n)$  the number  $x^T A x$  is less or equal to zero. Hence,  $W \cap U = \{0\}$  and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k. \quad (16)$$

This is the last inequality proving the statement.  $\square$

**Lemma 4.0.6.** Assume that  $\psi : U \rightarrow V$  is a chart and  $L : \mathbb{R}^m \rightarrow \mathbb{R}^m$  a linear function. Then the diagramm

$$\begin{array}{ccc} T_p M & \xrightarrow{q_\psi|_p} & \mathbb{R}^m \\ & \searrow & \downarrow L \\ & q_{L \circ \psi}|_p & \mathbb{R}^m \end{array}$$

commutes for all  $p \in U$ .

*Proof.* We show this by showing the inverse: For any  $v \in \mathbb{R}^m$  we have the two  $(q_\psi|_p)^{-1} \circ L^{-1}(v)$  and  $(q_{\psi \circ L}|_p)^{-1}$  and claim that they are the same. Assume that  $L^{-1} = (\lambda_{ij})_{ij}$ ,  $\psi(p) = x_0$ ,  $L(x_0) = \tilde{x}_0$ ,  $(v^1, \dots, v^m) = v \in \mathbb{R}^m$  and denote the basis of  $T_p M$  coming from the chart  $\psi$  by  $\left( \frac{\partial}{\partial x^i} \Big|_p \right)_i$  and the one coming from  $L^{-1} \circ \psi$  by  $\left( \frac{\partial}{\partial \tilde{x}^i} \Big|_p \right)_i$ . We now calculate

for  $f_p \in \mathcal{E}_p M$ :

$$\begin{aligned}
\left[ (q_{\psi \circ L}|_p)^{-1}(v) \right] (f_p) &= \sum_i v^i \frac{\partial}{\partial \tilde{x}^i} \Big|_p (f_p) \\
&= \sum_i v^i \frac{\partial}{\partial x^i} \Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v^i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x^j} \Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x^i} \Big|_{\tilde{x}_0} \\
&= \sum_{i,j} v^i \frac{\partial}{\partial x^j} \Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v^i q_{\psi}|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[ q_{\psi}|_p^{-1} \left( \sum_{i,j} v^i \lambda_{ji} e^j \right) \right] (f_p) \\
&= \left[ q_{\psi}|_p^{-1} (L^{-1}(v)) \right] (f_p)
\end{aligned}$$

This concludes the proof. □

**Theorem 4.0.7** (Simultaneous Diagonalisation of Quadratic Forms). *Let  $p$  be a critical point of a Morse function  $f$  on a manifold  $M$  and let  $g(p)$  be the Riemannian metric on  $T_p M$ . Then there exists a Morse chart  $\psi$  around  $p$  such that the representation of  $g(p)$  in the basis induced by the Morse chart is given by a diagonal matrix  $\text{diag}(\mu_1, \dots, \mu_m)$  with  $\mu_i > 0$  for all  $i$ .*

*Proof.* The quadratic form of  $f \circ \psi^{-1} - f(p)$  in the coordinates of any Morse chart  $\psi$  is  $q_H(v) = v^T H v$ . The quadratic form induced by the Riemannian metric  $g(p)$  is  $q_G(v) = v^T G v$ , where  $v \in \mathbb{R}^m$  are the coordinate vectors with respect to the Morse chart. To be precise this means for  $v, w \in \mathbb{R}^m$ :

$$g \circ (q_{\psi}|_p \oplus q_{\psi}|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v.$$

We now aim to manipulate  $\psi$  such that  $G$  becomes diagonal: Since  $g(p)$  is positive definite,  $G$  is also positive definite, and  $H$  is symmetric.

Consider the generalized eigenvalue problem  $Hv = \lambda Gv$ . Since  $H$  and  $G$  are real symmetric matrices and  $G$  is positive definite, this problem has  $m$  real eigenvalues  $\lambda_1, \dots, \lambda_m$  and corresponding eigenvectors  $v_1, \dots, v_m$ , which can be chosen to be orthogonal with respect to the bilinear form defined by  $G$ , such that  $v_i^T G v_j = \delta_{ij}$ , since if  $v_i$  and  $v_j$  are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (H v_j)^T v_i = \lambda_j (G v_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i).$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i) v_j^T G(v_i)}_{\neq 0} = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let  $L = [v_1 | \dots | v_m]$  be the matrix whose columns are these  $G$ -orthonormal eigenvectors. The linear change of coordinates  $y = Lz$  leads to new coordinates  $z$ . In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G (Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H (Lz) = z^T L^T H L z \end{aligned}$$

Since  $Hv_i = \lambda_i Gv_i$ , we have  $L^T H L = \text{diag}(\lambda_1, \dots, \lambda_m)$ . The eigenvalues  $\lambda_i$  are real and have the same signature as the eigenvalues of  $H$  ( $l$  negative,  $m-l$  positive). This is true to Sylvester's law of inertia. By a further scaling of the  $v_i$  and a reordering the matrix  $L^T H L$  can be brought into the form  $\text{diag}(-1, \dots, -1, 1, \dots, 1)$ . (by doing this however, the matrix  $L^T G L$  becomes  $L^T G L = \text{diag}(\mu_1, \dots, \mu_m)$  with  $\mu_j > 0 \ \forall j$ ). If  $\psi$  is the Morse chart we started with, then  $L^{-1} \circ \psi$  is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y)) = \sum_{i=1}^l -y_i^2 + \sum_{i=l+1}^m y_i^2$$

Furthermore, we want to inspect the metric  $g|_p : T_p M^2 \rightarrow \mathbb{R}$  with respect to the chart  $L^{-1} \circ \psi$ -induced basis  $\left( \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^m} \Big|_p \right)$ . By the lemma 4.0.6

$$g(q_{L^{-1} \circ \psi}^{-1} \Big|_p, q_{L^{-1} \circ \psi}^{-1} \Big|_p)(v, w) = g(q_\psi^{-1} \Big|_p, q_\psi^{-1} \Big|_p)(Lv, Lw) = (Lv)^T G L w = v^T L^T G L w$$

□

**Remark 4.0.8** (Linearly Approximate the Flow). Now we want to use this to Taylor expand the flow at the origin in said chart. We now want to linearly approximate  $\psi \circ \varphi_t \circ \psi^{-1} : U \rightarrow U$  which leads to the need of its Jacobian. It's not too hard to show that the flow has the Jacobian

$$q_\psi \frac{\partial}{\partial x} \Big|_0 \varphi_t(\psi^{-1}(x)) = \exp \left( \sum_k \left( g^{ki}(0) \frac{\partial^2 (f \circ \psi^{-1})}{\partial x^j \partial x^k} \right)_{ij} \right).$$

Now  $g^{ki}(0) = \delta_{ik} \mu_k$  with  $\mu_i > 0$  for all  $i$  and  $\frac{\partial^2 (f \circ \psi^{-1})}{\partial x^j \partial x^k} = s_k 2\delta_{jk}$  where  $s_k = -1$  if  $k \leq l$  and  $s_k = 1$  else. Hence:

$$\begin{aligned} \frac{\partial}{\partial x} \Big|_0 \psi \circ \varphi_t(\psi^{-1}(x)) &= -\exp(\text{diag}(-2\mu_1 t, \dots, -2\mu_l t, 2\mu_{l+1} t, \dots, 2\mu_m t)) \\ &= \text{diag}(e^{2\mu_1 t}, \dots, e^{2\mu_l t}, e^{-2\mu_{l+1} t}, \dots, e^{-2\mu_m t}) \end{aligned}$$

Hence we can linearly approximate the flow for fixed  $t$  in a chart centered at the critical point to get a function  $\mathcal{R} : V \rightarrow V$  such that  $\mathcal{R} \in \mathcal{O}(\|x\|^2)$ :

$$\begin{aligned}\psi \circ \varphi_t(\psi^{-1})(x) &= \psi \circ \varphi_t(\psi^{-1})(0) + \left( \frac{\partial}{\partial x} \Big|_0 \psi \circ \varphi_t(\psi^{-1}) \right) x + \mathcal{R}(x) \\ &= \left( \text{diag}(e^{2\mu_1 t}, \dots, e^{2\mu_l t}, e^{-2\mu_{l+1} t}, \dots, e^{-2\mu_m t}) \right) (x) + \mathcal{R}(x)\end{aligned}$$

We give the following definitions:

- $x^u := (x^1, \dots, x^l)$  and  $x^s := (x^{l+1}, \dots, x^m)$ , and with this we split  $x = (x^u, x^s)$ .
- $-(x^u)^2 := \sum_{i=1}^l -(x^i)^2$  and  $(x^s)^2 := \sum_{i=l+1}^m (x^i)^2$  and hence
$$f \circ \psi^{-1}(x^u, x^s) = a - (x^u)^2 + (x^s)^2.$$
- $e^{2\mu_u t} x^u := (e^{2\mu_1 t} x^1, \dots, e^{2\mu_l t} x^l)$  and similar  $e^{-2\mu_s t} x^s := (e^{-2\mu_{l+1} t} x^{l+1}, \dots, e^{-2\mu_m t} x^m)$ .
- $\mathcal{R}^u(x)$  and  $\mathcal{R}^s(x)$  such that  $\mathcal{R}(x) = (\mathcal{R}^u(x), \mathcal{R}^s(x))$

With these notations the calculation above yields:

$$\psi \circ \varphi_t(\psi^{-1})(x)(x^u, x^s) = \left( e^{2\mu_u t} x^u + \mathcal{R}^u(x), e^{-2\mu_s t} x^s + \mathcal{R}^s(x) \right).$$

Now the property of being in  $\psi(N_p \cap U)$  reads:

$$\begin{aligned}x \in V \text{ such that } & f(\varphi_T(\psi^{-1}(x))) && \geq a - \varepsilon \\ \Leftrightarrow & (f \circ \psi^{-1})(\psi \circ \varphi_T(\psi^{-1}(x))) && \geq a - \varepsilon \\ \Leftrightarrow & (f \circ \psi^{-1})(e^{2\mu_u T} x^u + \mathcal{R}^u(x), e^{-2\mu_s T} x^s + \mathcal{R}^s(x)) && \geq a - \varepsilon \\ \Leftrightarrow & -(e^{2\mu_u T} x^u + \mathcal{R}^u(x))^2 + (e^{-2\mu_s T} x^s + \mathcal{R}^s(x))^2 + \varepsilon && \geq 0\end{aligned}$$

For fixed  $x^s = a^s$  we inspect the slices and hope they are star-shaped: Assume that  $(a^u, a^s)$  satisfies the property:

$$\begin{aligned}-e^{4\mu_u T} a^u - 2e^{2\mu_u T} a^u \mathcal{R}^u(a^u, a^s) - \mathcal{R}^u(a^u, a^s)^2 + e^{4\mu_s T} a^s + 2e^{2\mu_s T} a^s \mathcal{R}^s(a^u, a^s) + \mathcal{R}^s(a^u, a^s)^2 \\ \geq -\varepsilon\end{aligned}$$

, then for any  $0 \leq \lambda \leq 1$ , the point  $(\lambda a^u, a^s)$  also satisfies the property: (here i am at the moment....) The next is just some ideas i tried but did not manage

We now inspect the function  $F : \mathbb{R}^m \rightarrow \mathbb{R}^l$  given by:

$$\begin{pmatrix} x^1 \\ \vdots \\ x^m \end{pmatrix} \mapsto \begin{pmatrix} -(e^{2\mu_1 T} x^1)^2 & +2e^{2\mu_1 T} x^1 \mathcal{R}^1(x) & +(\mathcal{R}^1(x))^2 & +(e^{-2\mu_s T} x^s + \mathcal{R}^s(x))^2 \\ \vdots & \vdots & \vdots & \\ -(e^{2\mu_l T} x^l)^2 & +2e^{2\mu_l T} x^l \mathcal{R}^l(x) & +(\mathcal{R}^l(x))^2 & \end{pmatrix}.$$

For this function we have that

$$\frac{\partial}{\partial x^s} F$$

is non-degenerat. To see this we calculate the partial differentials:

$$1. \frac{\partial}{\partial x^j} - (e^{2\mu_i T} x^i)^2$$